

Chapters 1 & 2 Review Worksheet

Basic Techniques for Solving Equations (§1.1–1.4) • Functions Review (§2.1–2.4)

Name _____

Date _____

Sections _____

Before you begin

- Start this set on graph or lined paper, or work directly on this handout. Put your **name**, the **date**, and the **book sections** at the top.
- Show every step. An answer with no supporting work earns no credit — the algebra *is* the assignment.
- Starred problems (★) are optional. Try them anyway; you will learn something even if you do not finish.
- When you are done, check against the solutions handout and make corrections in **blue pen**.

Part A — Isolation

§1.1

1. Solve for x : $\frac{2x-1}{3} - \frac{x+4}{5} = 2$
2. Solve for x : $\sqrt{3x+1} = x-3$. *Check each candidate in the original equation.*
3. Solve for x : $\frac{3}{x-2} = \frac{5}{x+4}$. State any values of x that must be excluded.
4. Solve for x : $|2x-5| = 7$
5. Solve for x in terms of a and b : $\frac{x}{a} + \frac{x}{b} = 1$. What must be true of a and b ?
6. The simple-interest formula is $A = P(1 + rt)$. Solve it for t .

Part B — Substitution & Elimination

§1.2–1.3

7. Solve by **substitution**: $y = 2x - 3$ and $3x + 2y = 8$

8. Solve by **elimination**: $4x - 3y = 18$ and $3x + 5y = -1$

9. Solve the system: $x + y + z = 6$, $2x - y + 3z = 9$, $x - 2y + 4z = 9$

10. Solve, or explain why no solution exists: $2x - 6y = 5$ and $-3x + 9y = 1$. *What do the two lines look like?*

11. Solve the system: $x^2 + y^2 = 25$ and $y = x + 1$. Describe the graphs and what the solutions mean.

12. Solve: $x + y = 5$, $y + z = 8$, $z + x = 9$. *Adding all three equations is faster than substituting.*

Part C — Lines & Graphing

§1.4

13. Find the equation of the line through $(-2, 5)$ and $(4, -7)$ in slope-intercept form, then give both intercepts.

14. Find the intersection of $2x + 3y = 12$ and $y = x - 1$ algebraically, then sketch both lines to confirm.



15. Consider the line $3x - 4y = 8$.
- Rewrite it in slope-intercept form and state the slope.
 - Find the line **parallel** to it through $(0, -1)$.
 - Find the line **perpendicular** to it through $(0, -1)$.
16. A theater sells 250 tickets and collects \$2,300. Adult tickets cost \$12 and student tickets cost \$7. How many of each were sold? *Define your variables before you write equations.*
17. A rectangle has perimeter 54 cm, and its length is 3 cm more than twice its width. Find its dimensions.

Part D — Function Basics

§2.1

18. Find the domain of each function. Write your answer in interval notation.
- $f(x) = \frac{x - 3}{x^2 - 4x - 5}$
 - $g(x) = \sqrt{7 - 2x}$
 - $h(x) = \frac{3}{\sqrt{x + 4}}$
19. Let $f(x) = 2x^2 - x + 1$. Compute and simplify $f(-3)$, $f(a + 1)$, and $f(-x)$.
20. Find the range of $f(x) = x^2 - 4x + 7$ by completing the square. Justify your answer.
21. For $f(x) = x^2 - 5x$, simplify the difference quotient $\frac{f(x + h) - f(x)}{h}$ ($h \neq 0$).

22. Which of these define y as a function of x ? Explain each answer.

a. $\{(1,2), (2,4), (3,2)\}$

b. $\{(1,2), (1,3), (2,5)\}$

c. $x = y^2$

d. $y = x^2$

Part E — Graphs of Functions

§2.2

23. Starting from $y = |x|$, list in order the transformations that produce $y = -2|x - 3| + 4$. Give the vertex, say which way it opens, and sketch it.



24. The graph of $y = f(x)$ passes through $(2, -5)$. Give a point that must lie on each graph below.

a. $y = f(x - 4) + 1$

b. $y = -3f(x)$

c. $y = f(2x)$

d. $y = f(-x)$

25. Sketch $y = \sqrt{x + 2} - 3$. State its domain, its range, and the coordinates of its endpoint.



26. Classify each as even, odd, or neither, and justify algebraically using $f(-x)$.

a. $f(x) = x^3 - 4x$

b. $g(x) = x^4 - 3x^2 + 1$

c. $h(x) = x^2 + x$

27. Let $f(x) = 2x - 1$ and $g(x) = x^2 + 3$. Find and simplify $f(g(x))$, $g(f(x))$, $f(g(2))$, and $g(f(2))$.
28. Let $f(x) = \frac{1}{x-2}$ and $g(x) = \sqrt{x}$. Find $(f \circ g)(x)$ and its domain. *Two separate restrictions apply.*
29. Write $h(x) = (3x + 1)^5$ as a composition $f(g(x))$ of two simpler functions, neither of which is x itself.
30. Let $f(x) = 2x - 3$. Find $f(f(x))$ and $f(f(f(x)))$. Predict the pattern for four applications.

Part G — Inverse Functions

31. Find $f^{-1}(x)$ for $f(x) = \frac{2x-5}{3}$, then verify that $f(f^{-1}(x)) = x$.
32. Find $f^{-1}(x)$ for $f(x) = 4 - x^3$.
33. Find $f^{-1}(x)$ for $f(x) = \frac{x+1}{x-3}$, and state the domain of f^{-1} .
34. Let $f(x) = x^2 - 6x + 5$ with domain restricted to $x \geq 3$. Find $f^{-1}(x)$ and its domain. *Complete the square first.*

35. Answer each in a sentence.

- a. f is invertible and $f(3) = 8$. What is $f^{-1}(8)$?
- b. What is $(f^{-1} \circ f)(5)$?
- c. Why does $g(x) = x^2$ have no inverse on all real numbers?
- d. How is the graph of f^{-1} related to the graph of f ?

Challenge Problems — optional



36. ★ Find all real solutions of $\sqrt{x+5} + \sqrt{x} = 5$.

37. ★ Let $f(x) = \frac{2x+3}{x-2}$. Show that $f(f(x)) = x$ for every x in the domain. What does that tell you about f^{-1} ?

38. ★ A function satisfies $f(x) + 2f\left(\frac{1}{x}\right) = 3x$ for every $x \neq 0$. Find $f(2)$, then find $f(x)$ in general.

39. ★ Find the range of $f(x) = \frac{2x+1}{x-3}$ without graphing. Solve $y = f(x)$ for x and ask which y are impossible.